

From Yes/No Questions to Information Gain*(work with a partner)*

All logarithms use base 2.

1. Identifying a mystery card

Your partner selects one of eight equally likely cards, numbered 1–8. You may ask only yes/no questions.

- (a) Write a first question that divides the possibilities into two equal groups.
- (b) Suppose the answer is “yes.” Write your next question.
- (c) With a good questioning strategy, how many questions are needed to identify the card? _____
- (d) Complete the table.

Equally likely cards	Questions needed to guarantee identification
2	
4	
8	
16	
32	

- (e) Explain how the number of questions changes when the number of cards doubles.
- (f) Write a formula relating the number of cards, N , to the number of questions, assuming N is a power of two.

2. Measuring information

The information associated with observing an outcome of probability p is

$$I(p) = -\log_2(p) = \log_2(1/p).$$

- (a) Complete the table.

Probability of the observed outcome	Information, in bits (How much information do I gain when I learn which outcome occurred?)
1	
1/2	
1/4	
1/8	

Information and Entropy*(work with a partner)***2. Measuring information (continued)**

- (b) Why does an outcome that was certain to occur give us zero information?
- (c) You flip two independent fair coins. What is the probability of observing heads on both? How much information does that outcome provide?
- (d) Each individual coin outcome provides one bit. Explain why your answer to (c) makes sense.

3. From information to entropy

A decision-tree node contains **eight blue examples and two orange examples**. We select an example uniformly at random.

- (a) What are the label probabilities?

$$P(\text{blue}) = \underline{\hspace{2cm}} \quad P(\text{orange}) = \underline{\hspace{2cm}}$$

- (b) Calculate the information provided by learning each possible label.

$$I(\text{blue}) = \underline{\hspace{2cm}} \quad I(\text{orange}) = \underline{\hspace{2cm}}$$

- (c) Which label provides more information when observed? Explain.

- (d) Entropy is the average information provided by learning the label:

$$H(Y) = - \sum_y p(y) \log_2 p(y).$$

Calculate the entropy of this node. (For a label with probability zero, its contribution to entropy is defined to be zero.)

- (e) A student computes $[I(\text{blue}) + I(\text{orange})]/2$. Why is this the wrong average for this node?
- (f) Rank these nodes from lowest to highest entropy. Explain without calculating.

Node	Blue examples	Orange examples
A	10	0
B	8	2
C	5	5

Converting features: continuous to discrete

Consider the following data table with one feature (temperature) and an output label ‘Y’ for playing tennis and ‘N’ for not playing tennis:

temp	label
80	Y
48	Y
60	N
48	N
40	N
48	Y
90	Y

Follow the steps below to convert this continuous feature to several discrete ones.

1. Sort examples based on the feature:
2. Different label with the same feature value? Collapse to “None”:
3. Whenever the label changes, make a feature (average adjacent feature values):
4. Write out the new data as it would look after feature conversion.